

AKASHDEEP

Full-Stack Engineer — AI & LLM Systems

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PROFESSIONAL SUMMARY

Full-Stack Engineer with 4.6+ years of experience designing, building, and deploying scalable backend and full-stack applications using Node.js, React.js, Python, FastAPI, and Express.js. Skilled in integrating AI/ML and LLM-based systems (RAG, LangChain, LangGraph, PyTorch) into production workflows, reducing manual effort by up to 35% and sustaining 99% uptime across live systems. Experienced in cloud-native development on AWS and Azure, CI/CD pipeline design, and Docker/Kubernetes containerization within Agile teams. Proven track record leading engineering teams, mentoring junior developers, and partnering directly with stakeholders to ship production-ready software on schedule.

TECHNICAL SKILLS

- **Languages:** JavaScript, TypeScript, Python, Java, SQL, HTML, CSS
- **Frontend:** React.js, Next.js, Bootstrap, TailwindCSS
- **Backend & Frameworks:** Node.js, Express.js, FastAPI, Python
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis
- **Cloud & DevOps:** AWS (EC2, S3, DynamoDB), Azure, Docker, CI/CD, Git, Linux
- **APIs & Architecture:** REST, GraphQL, Microservices Architecture, System & Software Architecture Design
- **AI & Machine Learning:** PyTorch, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases, LLM Integration

PROFESSIONAL EXPERIENCE

Full Stack Developer | Cosmic Owl Pvt Ltd, Bangalore, India

August 2024 – March 2026

- Built an AI Hybrid Production Tool and a Retrieval-Augmented Generation (RAG) AI Assistant using the Qwen2.5 LLM and a vector database, automating product specification generation and cutting manual effort by 30–35%.
- Optimized core API endpoints, improving response times by approximately 35% and sustaining 99% system uptime in production.
- Designed and deployed a computer vision pipeline (PyTorch, FastAPI) that automated product image-to-listing data conversion, reducing manual data entry by 35% and improving data accuracy.
- Implemented CI/CD pipelines and enforced software engineering best practices, reducing technical debt and accelerating deployment velocity.
- Led code reviews and mentored junior engineers, improving team knowledge-sharing and code quality.
- Partnered directly with stakeholders to align AI/ML solutions with business goals across the full software development lifecycle (SDLC).

Software Engineer | Napworks Pvt. Ltd., Mohali, India

July 2023 – July 2024

- Built Seatherny, an end-to-end avian species identification system combining an Android app with a TensorFlow deep learning model served via a Python pipeline on Google Cloud Platform (GCP).
- Optimized legacy systems, reducing latency and improving performance and reliability.
- Scaled cloud infrastructure to maintain 99% uptime across production deployments.
- Partnered with R&D and security teams to identify vulnerabilities and implement effective mitigation strategies.

Software Engineer | Techbit Solutions Pvt. Ltd., Chandigarh, India

September 2021 – December 2022

- Led a team of 3 engineers delivering full-stack and backend features across a portfolio of client products.

- Reduced API response times by approximately 35% through backend and server-side optimization.
- Owned client relationship management and delivery of critical modules, including analytics and billing systems.
- Mentored interns and junior engineers, improving onboarding efficiency and team capability.

Software Engineer | Bhanguz, Chandigarh, India

February 2021 – July 2021

- Engineered and deployed full-stack features using React.js on the frontend and Node.js/Python on the backend.
- Contributed to the design and development of a new software module, improving system efficiency and user experience.
- Collaborated with cross-functional teams to resolve technical issues and maintain seamless software operations.

KEY PROJECTS

RAG AI Assistant | Qwen2.5, LangChain, Vector DB, FastAPI, Redis

- Production retrieval-augmented generation assistant using the Qwen2.5 LLM and a vector database for semantic search over internal documentation, with caching and fallback handling for latency spikes. Result: ~35% faster response times, 99% uptime.

Product Listing Tool (Computer Vision) | Python, PyTorch, FastAPI, AWS

- Computer-vision-assisted pipeline that extracts product attributes from images and auto-drafts structured spec sheets for human review, replacing a fully manual process. Result: 30–35% reduction in manual effort.

Resume AI | React, Express.js, PostgreSQL, LLM API

- End-to-end resume analysis and tailoring tool: React frontend for upload and editing, Express.js backend handling parsing and LLM-driven suggestions, PostgreSQL for profiles and job-match history. Result: resume tailoring reduced from hours to minutes.

Seatherny | Android, TensorFlow, GCP

- End-to-end avian species identification app: Android frontend with an on-device TensorFlow model, served via a Python pipeline and hosted on Google Cloud Platform. Result: shipped to production with 99% uptime.

EDUCATION

Bachelor of Technology in Computer Science & Engineering (Coursework Complete)

Beant College of Engineering and Technology, Gurdaspur, India | 2019

ADDITIONAL EXPERIENCE

Trainee | CS Soft Solutions Pvt. Ltd., Chandigarh, India | January 2019 – June 2019